

DEDOUGOU, Burkina Faso

WMO: 655050

Lat: 12.467N

Lon: 3.483W

Elev: 300

StdP: 97.77

Time Zone: 0.00 (E00)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.5	18.7	-9.2	1.8	29.9	-7.2	2.1	29.6	7.4	27.6	6.8	27.6	1.6	40	0.380

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	11.1	40.6	20.6	39.8	20.5	39.1	20.5	26.7	32.0	26.2	31.8	25.9	31.6	2.8	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.4	21.4	29.1	24.9	20.7	28.7	24.5	20.2	28.5	85.4	31.9	83.5	31.8	82.0	31.5	31.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.4	6.4	5.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			7.4	27.8	0.8	0.5	6.8	28.1	6.3	28.4	5.9	28.6	5.3	29.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	29.0	25.7	28.7	32.0	33.3	32.5	30.0	27.6	26.5	27.3	29.5	29.1	26.3	(6)
(7)		DBStd	3.05	1.92	2.54	2.04	1.95	2.15	2.19	1.71	1.29	1.41	1.26	1.22	1.82	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	6949	488	522	683	699	698	601	546	512	518	604	573	504	(10)
(11)		CDD18.3	3907	229	289	425	449	440	351	287	254	268	346	323	246	(11)
(12)		CDH23.3	51721	2637	3908	6728	7227	6747	4774	3113	2302	2716	4318	4361	2891	(12)
(13)		CDH26.7	29221	1281	2252	4407	4866	4346	2623	1283	732	1111	2308	2546	1466	(13)
(14)	Wind	WSAvg	2.5	2.8	2.9	2.7	2.8	3.1	3.0	2.5	2.0	1.8	2.0	2.0	2.8	(14)
(15)	Precipitation	PrecAvg	808	0	1	6	23	74	104	190	228	129	43	2	0	(15)
(16)		PrecMax	1151	1	31	51	189	201	222	332	492	244	165	19	7	(16)
(17)		PrecMin	512	0	0	0	0	7	39	90	107	47	0	0	0	(17)
(18)		PrecStd	167	0	6	13	36	46	40	66	81	53	37	5	1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.7	39.6	41.3	42.0	41.5	39.0	35.7	32.8	34.8	37.8	38.1	36.6	(19)
MCWB			16.6	18.0	18.9	20.7	22.0	24.6	25.1	26.0	25.6	21.3	19.5	17.2	(20)	
2%		DB	34.9	38.3	40.2	40.7	40.2	37.5	34.1	31.9	33.7	36.8	37.3	35.4	(21)	
		MCWB	16.2	17.8	19.0	20.6	22.7	24.9	25.0	25.7	25.8	22.1	19.3	16.8	(22)	
5%		DB	33.6	37.2	39.5	39.9	39.1	36.2	32.9	31.1	32.7	35.9	36.6	34.3	(23)	
		MCWB	15.6	17.4	18.9	20.6	23.1	24.9	24.9	25.4	25.7	23.1	19.1	16.5	(24)	
10%		DB	32.4	35.9	38.6	39.1	37.9	35.0	31.9	30.2	31.7	34.9	35.8	33.2	(25)	
		MCWB	15.1	16.8	18.7	20.8	23.6	24.7	24.9	25.2	25.5	23.6	19.0	16.2	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.9	22.2	24.1	25.6	26.5	26.7	26.3	26.7	27.4	27.1	24.2	20.0	(27)
MCD B			28.6	33.0	35.0	34.6	34.4	33.5	31.2	31.1	32.3	31.9	32.6	31.9	(28)	
2%		WB	17.5	20.3	23.3	24.9	25.7	26.1	26.0	26.2	26.7	26.3	23.2	18.3	(29)	
		MCD B	32.6	32.9	34.0	34.5	34.3	33.3	31.4	30.5	31.6	31.8	31.3	32.8	(30)	
5%		WB	16.4	18.4	22.7	24.4	25.3	25.7	25.6	25.8	26.2	25.7	22.5	17.5	(31)	
		MCD B	32.3	35.1	33.6	34.4	33.9	32.9	30.8	29.9	31.1	31.6	30.8	32.4	(32)	
10%		WB	15.5	17.4	21.8	23.9	25.0	25.4	25.2	25.4	25.8	25.3	21.5	16.7	(33)	
		MCD B	31.4	34.8	33.6	34.2	33.6	32.6	30.2	29.3	30.5	31.3	30.8	31.7	(34)	
(35)	Mean Daily Temperature Range		MDBR	13.4	13.7	12.8	11.1	10.3	9.2	8.0	7.0	8.6	11.6	14.8	14.2	(35)
5% DB		MCD BR	15.2	15.1	13.7	12.3	11.5	10.5	9.2	8.2	10.0	13.0	15.3	15.4	(36)	
		MCWBR	5.7	5.6	5.3	4.6	3.3	2.7	2.8	3.2	3.7	3.6	5.0	5.3	(37)	
5% WB		MCD BR	14.6	14.1	10.9	10.5	10.0	9.5	8.5	7.8	9.4	10.8	13.0	15.1	(38)	
		MCWBR	6.0	6.0	5.3	3.9	3.2	3.3	3.3	3.4	4.1	3.9	4.5	5.5	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.546	0.604	0.725	0.716	0.681	0.643	0.572	0.539	0.536	0.583	0.535	0.508	(40)	
taud		1.837	1.706	1.506	1.611	1.658	1.755	1.949	2.036	2.025	1.882	1.887	1.928	(41)		
Ebn at Noon		757	732	658	662	672	689	744	780	787	740	761	778	(42)		
Edh at Noon		205	243	302	271	253	227	188	174	177	201	193	183	(43)		
(44)	All-Sky Solar Radiation	RadAvg	5.61	5.95	6.24	6.30	6.19	5.86	5.62	5.40	5.75	5.92	5.66	5.48	(44)	
(45)		RadStd	0.29	0.30	0.31	0.29	0.19	0.22	0.21	0.21	0.21	0.12	0.17	0.19	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page